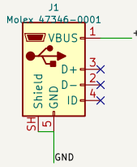
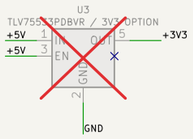
74ACT161 FREQUENCY DIVIDER — R4

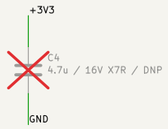
5 V DEFAULT | no output buffer | two opposing edge connectors

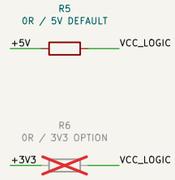
1. USB / J4 POWER AND OPTIONAL 3.3 V SUPPLY

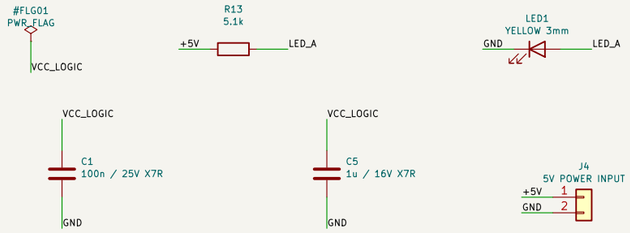












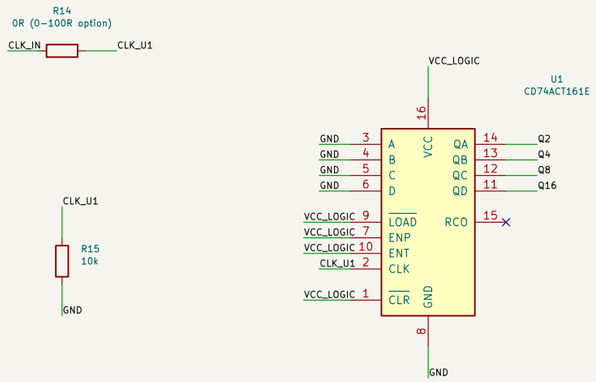
R5 fitted: VCC_LOGIC = input 5 V (default). R6/U3/C4 DNP.

3.3 V: REMOVE R5, fit R6/U3/C4 and REPLACE U1 with CD74AC161E.

Never power CD74ACT161E from 3.3 V. Never fit R5 and R6 together.

J4: regulated 5 V input. USB OR J4: use only one power source.

2. CLOCK AND COUNTER

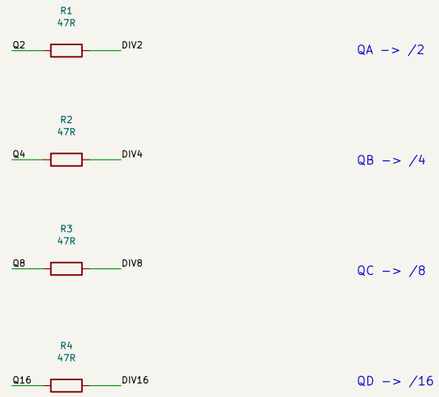


ACT / 5 V: TTL CLK; VIH >= 2 V, VIL <= 0.8 V; target up to 80 MHz.

AC / 3.3 V option: CLK 0..VCC; limit 64 MHz over −55..125 C.

No power-on reset: startup phase is undefined. C1 next to U1 pin 16.

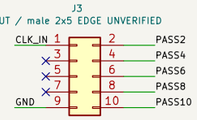
3. DIRECT OUTPUTS / SERIES DAMPING

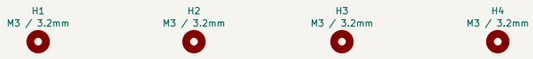


Output levels follow VCC_LOGIC. No SN74LVC125APW on this board.

4. CONNECTORS AND MOUNTING







PASS2/4/6/8/10 direct; no USB power on J2/J3. No 433 MHz support.

J2 LEFT EDGE, J3 RIGHT EDGE: 5 pads on F.Cu + 5 on B.Cu each.

Both mechanical interfaces UNVERIFIED. Confirm mating pin numbers.

Engineering prototype — UNVERIFIED edge connectors / analyzer

Sheet: /
File: ACT161_Frequency_Divider.kicad_sch

Title: 74ACT161 Frequency Divider

Size: A3	Date: 2026-10-09	**Rev: R4**
KiCad E.D.A. 10.0.6		Id: 1/1